

A SUPER-BAYESIAN'S SOLUTION TO A MULTI-BAYESIAN DECISION PROBLEM*

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Summary: A group of Bayesians with different priors has to take a joint decision. A super-Bayesian with his own prior and opinion of each Bayesian is called in by the group of Bayesians to assist them in their decision making. The Nash rule is used for reaching consensus between the Bayesians. The procedure is illustrated by an example.

1. INTRODUCTION

A decision problem that involves a super-Bayesian can be viewed from different angles, see Lindley, Tversky and Brown (1979) and Weerahandi and Zidek (1983). In these papers a super-Bayesian is assumed who would accumulate the opinions using standard Bayesian prior updating techniques. Lindley, Tversky and Brown develop two approaches involving an investigator or super-Bayesian. In the internal approach, the super-Bayesian estimates the Bayesian's "true" probabilities on the basis of

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his own assessments. In the external approach, the super-Bayesian updates his own probabilities in the light of the assessments made by the Bayesian.

In this paper a variation on the internal approach is considered where the super-Bayesian is called in to assist a group of Bayesians in reaching a joint decision. This means not only updating their probability assessments using his own knowledge, but also taking their respective utilities into account. We assume that the super-Bayesian himself is unaffected by the decision taken and thus has no utility function and is called in for his appreciation of the probabilities of the states of nature and his opinion of the Bayesians as probability assessors.

2. ELEMENTS OF THE DECISION PROBLEM

For simplicity, assume a parameter space with two points, i.e. $\Theta = \{\theta_1, \theta_2\}$. The decision space is $D = \{d_1, d_2, \dots, d_n\}$ and a group of k Bayesians $B_1, B_2, \dots, B_k$ has to take a joint decision from D .

2.1 The individual Bayesian's decision

Let B_i , $i=1,2,\dots,k$ denote the i -th Bayesian. In the absence of other persons, he would reach a personal decision using standard Bayesian techniques. The elements of this uni-Bayesian theory are summarized below.

(a) Bayesian i has his own prior information on θ .

$$P(\theta = \theta_1 | B_i) = w_i$$

$$P(\theta = \theta_2 | B_i) = 1 - w_i$$

- (b) Bayesian i has his own utilities.

B_i	d_1	d_2		d_n
θ_1	U_{i11}	U_{i21}		U_{in1}
θ_2	U_{i12}	U_{i22}		U_{in2}

Thus $U_{ij\ell}$ is the utility Bayesian i has for choosing decision d_j when the parameter is θ_ℓ $j=1,2,\dots,n$; $\ell=1,2$.

- (c) Bayesian i has his own observations $\underline{x}_i$ where

$$\underline{x}_i = [x_{i1}, x_{i2}, \dots, x_{in_i}].$$

- (d) The posterior distribution of Bayesian i can be found

$$P(\theta = \theta_1 | \underline{x}_i, B_i) = w_i^i$$

$$P(\theta = \theta_2 | \underline{x}_i, B_i) = 1 - w_i^i.$$

- (e) The expected utilities of Bayesian i , $U_i(d_j | w_i^i)$, can be calculated where $U_i(d_j | w_i^i)$ is the expected utility of Bayesian i for decision d_j .

$$\begin{aligned} U_i(d_j | w_i^i) &= U_{ij1} w_i^i + U_{ij2} (1 - w_i^i) \\ &= \frac{a_{ij} - b_{ij} w_i}{c_i - d_i w_i} \quad j=1,2,\dots,n \end{aligned}$$

where $a_{ij} = U_{ij2} P(\underline{x}_i | \theta = \theta_2)$

$$b_{ij} = a_{ij} - U_{ij1} P(\underline{x}_i | \theta = \theta_1)$$

$$c_i = P(\underline{x}_i | \theta = \theta_2)$$

$$d_i = c_i - P(\underline{x}_i | \theta = \theta_1).$$

- (f) Bayesian i can now obtain his Bayes decision by maximizing his expected utility.

2.2 The joint decision between the Bayesians using a super-Bayesian

The elements of the super-Bayesian's analysis are summarized as follows:

- (a) Suppose the super-Bayesian's prior on θ is $P(\theta) = p$ where he has some known probability distribution on p , say $\Pi(p)$. So for instance, $P(p=p_t) = \Pi(p_t) = p_t^i$, say, is known to the super-Bayesian.
- (b) The super-Bayesian has further his own opinion, expressed by the conditional density $\beta(w_i|p)$ of W_i conditional on the true value of p , about each of the Bayesians as probability assessors. The family of Beta distributions is natural for expressing an opinion on a probability, so assume that

$$w_i|p \sim \text{Beta}(a_i(p); b_i(p)), i=1,2,\dots,k.$$

Let $\mu_i(p)$ and $\sigma_i^2(p)$ denote the mean and variance respectively of the Beta distribution. The parameters $a_i(p)$ and $b_i(p)$ are chosen in such a way that

$$\mu_i(p) = p.$$

Hence

$$\frac{a_i(p)}{a_i(p) + b_i(p)} = p.$$

This means that the super-Bayesian assumes that the Bayesians are all equally unbiased in their assessment of p .

The restriction on the variance chosen by the super-Bayesian is

$$\sigma_i^2(p) < \mu_i(p)[1 - \mu_i(p)]$$

since

$$a_i(p) > 0 \quad \text{and} \quad b_i(p) > 0.$$

- (c) The conditional distribution $\Pi(p|w'_1, w'_2, \dots, w'_k)$ of the super-Bayesian on p given the Bayesians' posterior probabilities can be found where

$$P(p=p_t | w'_1, w'_2, \dots, w'_k) = \frac{\prod_{i=1}^k f(w'_i | p=p_t) P(p=p_t)}{\sum_j \prod_{i=1}^k f(w'_i | p=p_j) P(p=p_j)} = p_t$$

and $w_1, \dots, w_k$ are independent variables.

- (d) The expected utilities $U_i(d_j | w'_1, w'_2, \dots, w'_k)$ for each Bayesian i can be calculated by the super-Bayesian for each decision d_j . When calculating these expected utilities the posterior distribution of the super-Bayesian and his opinion of each of the Bayesians are taken into account. The utilities used in the calculation are those of the Bayesians since in the case considered in this paper they are the ones who will be affected by the decision. The formula for the altered expected utilities, as calculated by the super-Bayesian, is

$$\begin{aligned} U_i(d_j | w'_1, w'_2, \dots, w'_k) &= \sum_p \int_{w_i} U_i(d_j | w_i) \beta(w_i | p) dw_i P(p | w'_1, w'_2, \dots, w'_k) \\ &= \sum_p \int_{w_i} \frac{a_{ij} - b_{ij} w_i}{c_i - d_i w_i} \beta(w_i | p) dw_i P(p | w'_1, w'_2, \dots, w'_k) \\ &= \frac{b_{ij}}{d_i} + \left\{ \frac{a_{ij}}{c_i} - \frac{b_{ij}}{d_i} \right\} \sum_p {}_2F_1(1; a_i(p); a_i(p) \\ &\quad + b_i(p); \frac{d_i}{c_i}) P(p | w'_1, w'_2, \dots, w'_k) \\ &\quad i=1, 2, \dots, k \\ &\quad j=1, 2, \dots, n \end{aligned}$$

where ${}_2F_1$ is the hypergeometric series and is analytic if $|\frac{d_i}{c_i}| < 1$.

This gives, according to the super-Bayesian, the updated and more accurate estimate of the expected utilities for Bayesian i where he has incorporated his own opinion as well as the information yielded by the other Bayesians

This might mean a decision different from his optimal decision in the uni-Bayesian analysis for each Bayesian, but does not guarantee consensus among the k Bayesians. Intuitively it might be expected that consensus could be reached if the size of the observation set $\underline{x}_i$ is increased for all i , but according to Weerahandi and Zidek (1983) an example exists where opinions actually diverge as the amount of data increases.

A decision rule is thus still needed for reaching a joint decision. For this the Nash rule (Nash (1950)) can be used, which includes randomized solutions.

3. EXAMPLE

Assume two brothers, B_1 and B_2 , wish to go fishing. The brother B_2 is the elder. They can each, depending on the weather, make one of the following decisions.

- d_1 : Take the boat and go out on the sea
- d_2 : Do the fishing from the side
- d_3 : Stay at home.

Suppose the two brothers decide to ask their father (the super-Bayesian) to assist them in their decision making.

The states of nature are:

θ_1 : The weather is bad

θ_2 : The weather is good.

Bayesian 1

(a) Let the prior probabilities for Bayesian 1 be given by

$$P(\theta = \theta_1 | B_1) = w_1 = 0,7$$

$$P(\theta = \theta_2 | B_1) = 1 - w_1 = 0,3.$$

(b) Let his utilities be

B_1	d_1	d_2	d_3
θ_1	0	2	10
θ_2	4	8	9

(c) Assume the random variable X_1 can be observed by Bayesian 1 where

$X_1 = 1$ bad weather is predicted

$X_1 = 0$ good weather is predicted,

and where it is known that

$$P(X_1 = 0 | \theta = \theta_1) = 0,5$$

$$P(X_1 = 0 | \theta = \theta_2) = 0,6.$$

Suppose $X_1 = 0$ has been observed.

(d) The posterior probability for Bayesian 1, given that $X_1 = 0$ has been observed, is given by

$$P(\theta = \theta_1 | B_1, X_1 = 0) = \frac{0,5w_1}{0,6 - 0,1w_1} = w_1' = 0,6604.$$

(e) The expected utilities are $U_1(d_j|w_1^1) = \frac{a_{1j} - b_{1j}w_1}{c_1 - d_1w_1}$ $j=1,2,3$

which yields $U_1(d_1|w_1^1) = 1,3585$

$U_1(d_2|w_1^1) = 4,0377$

$U_1(d_3|w_1^1) = 9,6604.$

(f) Hence the Bayes decision is d_3 for Bayesian 1.

Bayesian 2

Assume the following holds for Bayesian 2.

(a) Prior probabilities: $P(\theta = \theta_1|B_2) = w_2 = 0,4$

$P(\theta = \theta_2|B_2) = 1 - w_2 = 0,6.$

(b) Utilities:

B_2	d_1	d_2	d_3
θ_1	4	7	8
θ_2	10	9	4

(c) Assume Bayesian 2 can observe X_2 where

$X_2 = 1$ bad weather is predicted

$X_2 = 0$ good weather is predicted

and it is known that

$P(X_2 = 0|\theta = \theta_1) = 0,3$

$P(X_2 = 0|\theta = \theta_2) = 0,8.$

Suppose $X_2 = 0$ has been observed.

- (d) Hence the posterior probability for Bayesian 2, given that $X_2 = 0$ has been observed, is given by

$$P(\theta = \theta_1 | B_2, X_2 = 0) = \frac{0,3w_2}{0,8 - 0,5w_2} = w_2' = 0,2.$$

- (e) The expected utilities are $U_2(d_j | w_2') = \frac{a_{2j} - b_{2j}w_2}{c_2 - d_2w_2}$ $j=1,2,3$

$$\text{which yields } U_2(d_1 | w_2') = 8,8000$$

$$U_2(d_2 | w_2') = 8,6000$$

$$U_2(d_3 | w_2') = 4,8000$$

- (f) Therefore the Bayes decision is d_1 for Bayesian 2.

The super-Bayesian (N)

We shall now consider the problem from the super-Bayesian's point of view.

- (a) Let his prior probabilities be:

$$P(\theta = \theta_1 | N) = p$$

$$P(\theta = \theta_2 | N) = 1-p.$$

Assume N has the following probability distribution on p :

$$P(p = 0,2) = 0,5$$

$$P(p = 0,4) = 0,3$$

$$P(p = 0,6) = 0,2.$$

- (b) Let W_i be the random variable from which Bayesian i 's prior probability $P(\theta = \theta_1 | B_i) = w_i$ is a certain value, $i=1,2$. The super-Bayesian's opinion of the two Bayesians is reflected in a probability distribution he has on W_1 and W_2 separately. Assume that he has the following opinions of the Bayesians:

	Mean	Variance
$w_1 p = 0,2 \sim \text{Beta}(0,0286 ; 0,1143)$	0,2	0,14
$w_1 p = 0,4 \sim \text{Beta}(0,08 ; 0,12)$	0,4	0,2
$w_1 p = 0,6 \sim \text{Beta}(0,0546 ; 0,0364)$	0,6	0,22
$w_2 p = 0,2 \sim \text{Beta}(0,12 ; 0,48)$	0,2	0,1
$w_2 p = 0,4 \sim \text{Beta}(0,4727 ; 0,7091)$	0,4	0,11
$w_2 p = 0,6 \sim \text{Beta}(0,4286 ; 0,2857)$	0,6	0,14

- (c) Given the two Bayesians' posterior probabilities on θ , namely w_1^i and w_2^i , he can calculate conditional probabilities for p , namely:

$$P(p = 0,2 | w_1^i, w_2^i) = 0,2507$$

$$P(p = 0,4 | w_1^i, w_2^i) = 0,6245$$

$$P(p = 0,6 | w_1^i, w_2^i) = 0,1248.$$

- (d) Bayesian N can now calculate the following expected utilities for the two Bayesians:

$$B_1: U_1(d_1 | w_1^i, w_2^i) = 2,5244$$

$$U_1(d_2 | w_1^i, w_2^i) = 5,7866$$

$$U_1(d_3 | w_1^i, w_2^i) = 9,3689$$

$$B_2: U_2(d_1 | w_1^i, w_2^i) = 8,1765$$

$$U_2(d_2 | w_1^i, w_2^i) = 8,4530$$

$$U_2(d_3 | w_1^i, w_2^i) = 5,0941.$$

The Bayes decision for B_1 is, according to the super-Bayesian, d_3 .

The Bayes decision for B_2 is, according to the super-Bayesian, d_2 .

The Nash solution

The Bayesians have to randomize between d_2 and d_3 with

$$\alpha_1 \frac{U_2(d_2|w_1^1, w_2^1)[U_1(d_3|w_1^1, w_2^1) - U_1(d_2|w_1^1, w_2^1)] + \alpha_2 U_1(d_2|w_1^1, w_2^1)[U_2(d_3|w_1^1, w_2^1) - U_2(d_2|w_1^1, w_2^1)]}{[U_2(d_2|w_1^1, w_2^1) - U_2(d_3|w_1^1, w_2^1)][U_1(d_3|w_1^1, w_2^1) - U_1(d_2|w_1^1, w_2^1)]}$$

where δ_0 is the probability of choosing decision d_3 (see De Waal (1982)), and α_i is the weight given to Bayesian i , $\sum_i \alpha_i = 1$.

Suppose the super-Bayesian considers the elder brother, B_2 , the more responsible of the two, so that his preference should carry a little more weight, thus let $\alpha_1 = 0.4$ and $\alpha_2 = 0.6$.

Generally, the decision on the weights need not be the super-Bayesian's responsibility. This can be agreed upon among the Bayesians themselves if any weights other than equal ones are used.

With the weights as given above, (2.1) yields $\delta_0 = 0.0375$.

The Nash solution is then:

Choose d_3 with a probability of 0.0375
 Choose d_2 with a probability of 0.9625.

This solution gives expected utilities of

$$U_1(\delta_0 d_3 + (1 - \delta_0) d_2 | w_1^1, w_2^1) = 5.9209$$

$$\text{and } U_2(\delta_0 d_3 + (1 - \delta_0) d_2 | w_1^1, w_2^1) = 8.3270.$$

Suppose the two Bayesians did not use the super-Bayesian in their decision making. Then the Bayesians have to randomize between d_1 and d_3 .

Let $\alpha_1 = 0,4$ and $\alpha_2 = 0,6$, then $\delta_0 = 0,7818$ with the Nash solution;

choose d_3 with a probability of 0,7818

choose d_1 with a probability of 0,2182

and expected utilities

$$U_1(\delta_0 d_3 + (1 - \delta_0) d_1 | w_1') = 7,8489$$

$$U_2(\delta_0 d_3 + (1 - \delta_0) d_1 | w_2') = 5,6728$$

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